

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title: COMPLETED RESEARCH		Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 10

(1) Research Program/Project/Study Title: <i>Growth and Yield of Potted Pechay as Influenced by Different Media Combinations Containing Mushroom Spent Compost and Cropping Season</i>	
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Program or Project has component (*if no proceed to 2*)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Mambusao
		Province:	Capiz
		Region:	VI
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Vegetable		(8) Discipline/Field of Specialization Agriculture	

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 2 of 10

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted at the Organic Agriculture project of Capiz State University, Burias, Mambusao, Capiz, to determine the effect of different media combinations containing mushroom spent compost on the production performance of potted pechay in two cropping seasons. The study was a two-factor experiment laid out in Completely Randomized Design with three replications. Factor A was the different cropping period which comprised of A1- first cropping (November - December), and A2- second cropping (March - April). Factor B was the different growing media such as B1- Pure mushroom spent, B2- MS + GS (1:1), B3- MS + PDCRH (1:1), B4- MS + GS + PDCRH (1:1:1), B5 - Compost + GS + Sand (1:1:1) and B6 - Pure Garden Soil. The data gathered included the growth and yield of pechay were subjected to the analysis of variance using STAR software, LSD test for mean comparison and DMRT for interaction effect.

The result of the study showed that all the growth and yield parameters of potted pechay were not affected by the different media combinations containing mushroom spent compost. However, parameters such as plant height, stem girth, width of leaves, length of longest leaves, weight of roots, length of longest roots and weight of non-marketable plant and marketable plant were highly affected by the cropping season. There was significant interaction effect noted between the cropping seasons and different media combination on the length of longest leaves.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

organic farming, media combinations, mushroom spent, compost, potted pechay

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

The utilization of agricultural waste and by products in sustainable agriculture has gained significant attention recently. One promising agricultural practice involves using mushroom spent compost, which is a nutrient-rich organic substrate produced during mushroom cultivation, as a soil conditioner and fertilizer. This research aims to explore the potential impact of mushroom spent compost on the growth and yield of Pechay (*Brassica chinensis* Linn), a leafy green vegetable with significant agricultural and economic importance.

Numerous studies have emphasized the positive effects of organic amendments, such as compost, on crop growth and yield. Compost can improve soil structure, nutrient levels, and microbial activity, ultimately enhancing plant health and productivity (Liu et al., 2015; Bittman et al., 2011). However, the specific effects of mushroom spent compost, a unique organic material, on pechay have not been extensively investigated.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 3 of 10

The concept of repurposing agricultural waste products like mushroom spent compost aligns with the principles of sustainable agriculture, contributing to soil fertility and reducing the environmental burden of waste disposal (Jenkins et al., 2004). Research in this field can offer valuable insights into how to harness the potential of agricultural waste for the betterment of crop production.

Finally, understanding the influence of mushroom spent compost on pechay can have broader implications for sustainable vegetable production. Pechay is widely cultivated and consumed in various parts of the world, making it a crucial crop for both small-scale and large-scale agriculture (Gupta et al., 2016). Therefore, assessing the impact of mushroom spent compost on its growth and yield could provide practical guidance for farmers and agricultural practitioners.

Objectives of the Study

The study was conducted to determine the growth and yield performance of potted pechay grown in different media combinations containing mushroom spent compost and cropping seasons. Specifically, the study aims to:

1. Determine the growth and yield performance of potted pechay as affected by the different media combinations containing mushroom spent compost;
2. Determine the effect of cropping seasons on the growth and yield of potted pechay;
3. Evaluate which media and cropping seasons will give the best growth and yield to potted pechay; and
4. Determine the interaction effect between the different media combinations containing mushroom spent compost and cropping seasons.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The study was conducted at the Organic Agriculture project of Capiz State University- Burias Campus, Burias, Mambusao, Capiz. The study was a two-factor experiment laid out in Completely Randomized Design with three replications. Factor A was the different cropping season which comprised A1- first cropping (November - December), and A2- second cropping (March - April). Factor B was the different media combination containing mushroom spent compost such as B1- Mushroom spent (MS), B2- MS + Garden Soil (GS) (1:1), B3- MS + Partially Decompose Ricehull (PDCRH) (1:1), B4- MS + GS + PDCRH (1:1:1), B5 - Compost + GS + Sand (1:1:1) and B6 - Garden Soil. Growth and yield parameters of potted pechay were subjected to the Analysis of Variance (ANOVA) using STAR software, LSD test was used for mean comparison and DMRT for the interaction effect.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 4 of 10

figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Plant height (cm)

The analysis of variance for the plant height of pechay revealed a highly significant effect of cropping seasons. This means that the cropping season highly affected the height of pechay. In terms of different media combinations containing mushroom spent compost, it does not have significant effect to the height of pechay.

Pechay grown in the first cropping season produced the tallest plant height with the mean of 19.72 cm while pechay grown during the second cropping has the shortest plant height with the mean of 17.73 cm.

No interaction effect between the two factors was reflected in the result of ANOVA. This implies that the two factor did not interact in influencing the plant height of pechay.

Stem girth (mm)

The ANOVA for the stem girth of pechay showed that cropping season highly affected the stem girth of pechay while the different media combinations containing mushroom spent compost revealed that it does not significantly affect the stem girth. Mean ranged from 6.16 mm to 6.93 mm. No interaction between the two factors was reflected in the result of ANOVA.

Pechay grown during the second cropping produced the biggest stem girth with a mean of 7.18 mm, while pechay grown on the first cropping produced the smallest stem girth (5.89 mm).

There was no interaction effect between the two factors in influencing the stem girth of pechay.

Number of Leaves

The number of leaves of pechay was not affected both by cropping and different media combinations containing mushroom spent compost as revealed in the result of ANOVA. For the different growing media the mean ranged from 7.12 to 7.90 and for cropping season the mean ranged from 7.33 to 7.82, respectively.

There was no interaction effect between two factors in influencing the number of leaves of pechay.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 10

Width of Leaves (cm)

Analysis of Variance revealed that the width of leaves was significantly affected by cropping but not with different media combinations containing mushroom spent compost. Mean ranged from 8.09 mm to 8.91 mm among the different media.

Pechay grown on the first cropping produced bigger leaves (8.99 cm), compared to those pechay grown during second cropping (8.12 cm).

There was no interaction effect noted between different media combination containing mushroom spent compost and cropping seasons.

Length of longest Leaves (cm)

The analysis of variance for the length of the longest leaves revealed that the cropping season significantly affects the length of the longest leaves of pechay. Likewise, an interaction effect between the two factors was also reflected in the result of ANOVA. This implies that the two factors interacted in influencing the length of leaves of pechay.

During the first cropping, pechay grown in a media composed of Compost, GS, and Sand produced the longest leaves with a mean of 20.29 cm, which significantly differed from pechay grown in other media combinations such as MS, GS, and PDCRH (18.34 cm), MS (17.79 cm), MS and PDCRH (17.68 cm), MS and GS (17.37 cm) and GS (16.84 cm) grown in the same cropping season.

The shortest leaves were recorded from pechay grown in MS with a mean of 10.42 cm during the second cropping, which is found to be comparable with those grown in media composed of Compost, GS and Sand (10.91 cm), MS and GS (11.03 cm), MS with PDCRH (11.46 cm), GS (12.01 cm), and MS, GS PDCRH (12.12 cm) grown during the same cropping season.

The result implied that ideal potting media can be used regardless of the cropping season, however, media containing mushroom spent may be utilized during the first cropping.

Biomass (g)

Analysis of variance for the biomass revealed a not significant result both for the cropping season and the different media combinations containing mushroom spent compost. For the cropping season, the mean ranged from 22.95 g to 24.33 g. The mean of different media combinations ranged from 17.73 g to 29.83 g.

There was no significant interaction effect between cropping seasons and

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 6 of 10

media combinations containing mushroom spent compost reflected in the result of ANOVA.

Weight of marketable plant (g)

The weight of marketable pechay significantly differed with the cropping seasons but not with the different growing media. For media combinations containing mushroom spent compost means weight of marketable pechay ranged from 15.3 g to 29.83 g.

The LSD revealed that the marketable pechay harvested in the first cropping obtained a heavier weight (21.71 g) compared to those pechay plants grown during the second cropping (15.56 g).

Analysis of Variance revealed no interaction effect between the two factors in influencing the weight of marketable pechay.

Weight of non-marketable Plant (g)

The ANOVA revealed that the weight of non-marketable pechay was affected by cropping seasons but not with the different media combinations containing mushroom spent compost. The mean weight of non-marketable pechay as affected by the different media combinations ranged from 3.06 g to 4.87 g.

The LSD revealed that the plants harvested in the second cropping obtained heavier non-marketable pechay (4.62 g) compared to those produced during the first cropping seasons (2.64 g).

There was no interaction effect between the two factors in affecting the weight of non-marketable plants.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

CONCLUSION

Based on the result of the study the following conclusion was drawn:

1. The growth and yield of potted pechay were not significantly affected by the different combinations of media containing mushroom spent compost.
2. Cropping season significantly affected the growth and yield parameters of potted pechay, with the exception of the number of leaves and biomass.
3. Potting media combinations containing Mushroom Spent and Garden Soil can

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 7 of 10

ideally be used regardless of the cropping season.

4. An interaction was found between media combinations containing mushroom spent compost and cropping season in relation to the length of the longest leaves of potted pechay.

RECOMMENDATION

1. Use potting media containing Mushroom spent and Garden soil to potted pechay.
2. Potted pechay is ideally planted during the months of November to December.
3. Use any potting media combinations whether it is wet or dry season in growing pechay. However, media containing mushroom sent may be utilized during the first cropping
4. Conduct similar study to test the effectiveness of other media combinations containing different compost materials to other vegetables.
5. Conduct similar study to test the effectiveness of different compost material combinations during other wet and dry months of the year.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

Liu, X., Li, X., Li, M., & Tang, C. (2015). Effects of organic manure and organic-inorganic compound fertilizer on crop yield and soil organic matter content. *Soil Science and Plant Nutrition*, 61(2), 313-322.

Bittman, S., Hunt, D. E., & Topp, E. (2011). The potential to mitigate greenhouse gas emissions in intensive agriculture by substituting synthetic N fertilizers with anaerobically digested pig slurry. *Agriculture, Ecosystems & Environment*, 141(3-4), 359-372.

Jenkins, W. R., Aerts, R., Mulligan, D. R., & Neilson, R. (2004). Soil carbon and nitrogen storage in reforested land: A 30-year ecosystem perspective. *Ecological Applications*, 14(2), 213-127.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 8 of 10

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	Document Type: FORM ISO 9001:2015	Document Code	RES-F08
		Revision No.	00
	Document Title: COMPLETED RESEARCH	Effective Date	January 26, 2023
		Page	Page 1 of 10

Appendices

TREATMENT	Plant Height (cm)		Stem girth (mm)		Number of leaves ^{ns}		Width of Leaves(cm)		Length of leaves(cm)**		Biomass(g) ^{ns}	
	Cropping**		Cropping**		Cropping		Cropping*		Cropping		Cropping	
	1	2	1	2	1	2	1	2	1	2	1	2
A - Pure mushroom spent	18.87	17.49	5.83	6.49	8.1	7.4	8.45	70.72	17.79 b	10.42 c	24.23	18.6
B - MS + GS (1:1)	19.38	17.69	5.7	6.64	7.13	7.9	8.53	7.94	17.37 b	11.03 c	21.23	21.8
C - MS + PDCRH (1:1)	19.17	17.82	5.97	7.88	7.03	7.6	9.26	8.35	17.68 b	11.46 c	24.3	26
D - MS + GS + PDCRH (1:1:1)	20.16	18.79	5.97	7.81	7.6	8.1	8.73	8.85	18.34 b	12.12 c	23	29.83
E - Compost + GS + Sand (1:1:1)	22.16	16.37	5.87	6.92	8.23	7.57	10.33	7.49	20.29 a	10.91 c	32.7	17.73
F - Pure Garden Soil	18.6	18.19	6.03	7.33	5.87	8.37	8.65	8.33	16.84 b	12.01 c	20.5	23.73
Total	118.34	106.35	35.37	43.07	43.96	46.94	53.95	111.68	108.31	67.95	145.96	137.69
Mean	19.72a	17.73b	5.89b	7.18a	7.33	7.82	8.99a	8.12b	18.05a	11.33b	24.33	22.95

Table 1. Growth parameters of potted pechay as affected by different media combinations containing mushroom spent compost and cropping seasons.

	Document Type: FORM ISO 9001:2015	Document Code	RES-F08
		Revision No.	00
	Document Title: COMPLETED RESEARCH	Effective Date	January 26, 2023
		Page	Page 2 of 10

TREATMENT	Weight of Marketable Part(g)		Weight of Non- Marketable Part(g)	
	Cropping*		Cropping**	
	1	2	1	2
A - Pure mushroom spent	20.77	12.3	2.28	3.83
B - MS + GS (1:1)	16.67	13.93	2.83	6.9
C - MS + PDCRH (1:1)	30.97	18.43	2.21	3.93
D - MS + GS + PDCRH	18.37	19.77	2.86	5.77
E - Compost + GS + Sand (1:1:1)	27.7	12.83	3.25	2.9
F - Pure Garden Soil	15.8	16.1	2.43	4.4
Total	130.28	93.36	15.86	27.73
Mean	21.71a	15.56b	2.64b	4.62a

Table 2. Yield parameters of pechay potted as affected by different media combinations containing mushroom spent compost and cropping seasons.